

## CSE 321 Introduction to Algorithm Design Quiz 1 Solution

### Question 1

(50 points)

Compare the following functions in terms of their asymptotic growth rate. Order them from the slowest-growing to the fastest-growing function. Provide mathematical reasoning for your ordering.

$$f_1(n) = \log(n!)$$

$$f_2(n) = n2^n$$

$$f_3(n) = 5^{\log_2 n}$$

$$f_4(n) = n \log n$$

$$f_5(n) = n(\log n)^{0.5}$$

If necessary, apply Stirling's approximation for factorial expressions:

$$\log(n!) \approx n \log n - n$$

**Note:**  $f(n) < g(n)$  means  $f(n)$  grows asymptotically slower than  $g(n)$  and  $f(n) \sim g(n)$  means the growth rate of  $f(n)$  and  $g(n)$  are similar.

**(Solution)**

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{f_1(n)}{f_2(n)} &= \lim_{x \rightarrow \infty} \frac{\log(n!)}{n \cdot 2^n} \\ &= \lim_{x \rightarrow \infty} \frac{n \log n - n}{n \cdot 2^n} && \text{-by Stirling's approximation} \\ &= \lim_{x \rightarrow \infty} \frac{n(\log n - 1)}{n \cdot 2^n} \\ &= \lim_{x \rightarrow \infty} \frac{\log n - 1}{2^n} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{1}{n}}{2^n \cdot \ln 2} && \text{-by L'Hospital's rule} \\ &= \lim_{x \rightarrow \infty} \frac{1}{n \cdot 2^n \cdot \ln 2} \\ &= \lim_{x \rightarrow \infty} \frac{1}{n \cdot 2^n \cdot \ln 2} \\ &= 0 \\ &\Rightarrow f_1(n) = o(f_2(n)) \\ &\Rightarrow f_1(n) < f_2(n) \end{aligned}$$

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$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{f_1(n)}{f_3(n)} &= \lim_{x \rightarrow \infty} \frac{\log(n!)}{5^{\log_2 n}} \\
 &= \lim_{x \rightarrow \infty} \frac{n \log n - n}{5^{\log_2 n}} && \text{-by Stirling's approximation} \\
 &= \lim_{x \rightarrow \infty} \frac{n \log n - n}{n^{\log_{25}}} \\
 &= \lim_{x \rightarrow \infty} \frac{n \cdot (\log n - 1)}{n^{\log_{25}}} \\
 &= \lim_{x \rightarrow \infty} \frac{\log n - 1}{n^{\log_{25} - 1}} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{1}{n}}{(\log_{25} - 1) \cdot n^{\log_{25} - 2}} && \text{-by L'Hospital's rule} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{(\log_{25} - 1) \cdot n^{\log_{25} - 1}} \\
 &= 0
 \end{aligned}$$

$$\Rightarrow f_1(n) = o(f_3(n))$$

$$\Rightarrow f_1(n) < f_3(n)$$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{f_3(n)}{f_2(n)} &= \lim_{x \rightarrow \infty} \frac{5^{\log_2 n}}{n \cdot 2^n} \\
 &= \lim_{x \rightarrow \infty} \frac{n^{\log_{25}}}{n \cdot 2^n} \\
 &= \lim_{x \rightarrow \infty} \frac{n^{\log_{25} - 1}}{2^n} \\
 &= \lim_{x \rightarrow \infty} \frac{(\log_{25} - 1) \cdot n^{\log_{25} - 2}}{2^n \cdot \ln 2} && \text{-by L'Hospital's rule} \\
 &= \lim_{x \rightarrow \infty} \frac{(\log_{25} - 1) \cdot (\log_{25} - 2) \cdot n^{\log_{25} - 3}}{2^n \cdot \ln^{22}} && \text{-by L'Hospital's rule} \\
 &= \lim_{x \rightarrow \infty} \frac{(\log_{25} - 1) \cdot (\log_{25} - 2)}{n^{3 - \log_{25}} \cdot 2^n \cdot \ln^{22}} && \text{-since } \log_{25} - 3 < 0 \\
 &= 0
 \end{aligned}$$

$$\Rightarrow f_3(n) = o(f_2(n))$$

$$\Rightarrow f_3(n) < f_2(n)$$

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$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{f_1(n)}{f_4(n)} &= \lim_{x \rightarrow \infty} \frac{\log(n!)}{n \log n} \\ &= \lim_{x \rightarrow \infty} \frac{n \log n - n}{n \log n} && \text{-by Stirling's approximation} \\ &= \lim_{x \rightarrow \infty} \frac{n \log n}{n \log n} - \lim_{x \rightarrow \infty} \frac{n}{n \log n} \\ &= 1 - 0 \\ &= 1\end{aligned}$$

$$\Rightarrow f_1(n) = \theta(f_4(n))$$

$$\Rightarrow f_1(n) \sim f_4(n)$$

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{f_1(n)}{f_5(n)} &= \lim_{x \rightarrow \infty} \frac{\log(n!)}{n(\log n)^{0.5}} \\ &= \lim_{x \rightarrow \infty} \frac{n \log n - n}{n(\log n)^{0.5}} && \text{-by Stirling's approximation} \\ &= \lim_{x \rightarrow \infty} \frac{n(\log n - 1)}{n(\log n)^{0.5}} \\ &= \lim_{x \rightarrow \infty} \frac{\log n - 1}{(\log n)^{0.5}} \\ &= \lim_{x \rightarrow \infty} \frac{\log n}{(\log n)^{0.5}} - \lim_{x \rightarrow \infty} \frac{1}{(\log n)^{0.5}} \\ &= \lim_{x \rightarrow \infty} (\log n)^{0.5} - 0 \\ &= \infty\end{aligned}$$

$$\Rightarrow f_1(n) = \omega(f_5(n))$$

$$\Rightarrow f_5(n) < f_1(n)$$

**Therefore,**

$$f_5(n) < f_4(n) \sim f_1(n) < f_3(n) < f_2(n)$$

## CSE 321 Introduction to Algorithm Design Quiz 1 Solution

### Question 2

(50 points)

**(a) Prove that  $f(n) = 3^n$  belongs to  $\Omega(2^n)$ , i.e., show that:**

$$3^n \in \Omega(2^n)$$

**(b) Prove that  $n^2$  does not belong to  $O(n)$ , i.e., show that:**

$$n^2 \notin O(n)$$

#### **(Solution)**

**(a)** Does there exist some positive constants  $c$  and  $n_0$  such that  $3^n \geq c \cdot 2^n$  for all  $n \geq n_0$ ?

Let  $c = 1$  and  $n_0 = 1$ . Then prove if  $3^n \geq 2^n$  for all  $n \geq 1$  by induction.

For  $n = 1$ ,  $3^1 = 3 \geq 2 = 2^1$ . (base step)

For  $n = k > 1$ , assume  $3^k \geq 2^k$ . (inductive hypothesis)

For  $n = k + 1$ ,

$$\begin{aligned} 3^{k+1} &= 3^k \cdot 3^1 \\ &\geq 2^k \cdot 2^1 && \text{-by inductive hypothesis and base step} \\ &\geq 2^{k+1} \end{aligned}$$

**Therefore**,  $3^n \geq c \cdot 2^n$  for all  $n \geq n_0$  where  $c = 1$  and  $n_0 = 1$ . That means  $3^n \in \Omega(2^n)$ .

**(b)** Assume  $n^2 \in O(n)$ , meaning there are some positive integers  $c$  and  $n_0$  such that  $n^2 \leq c \cdot n$  for all  $n \geq n_0$ . However, if we divide both sides of the inequality  $n^2 \leq c \cdot n$  by  $n$ , which is positive for all  $n \geq n_0$ , we get:

$$n \leq c$$

This implies that for all  $n \geq n_0$ , the constant  $c$  must be greater than or equal to  $n$ . But this leads to a contradiction, since  $n$  can grow arbitrarily large, while  $c$  is a fixed constant.

**Therefore**, the initial assumption is false, and we conclude that:

$$n^2 \notin O(n)$$